

Breaking Self-Verification Bias in TableQA: An Iterative Multi-Agent Framework via Adversarial Critique Distillation

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Abstract. Table Question Answering (TableQA) stands as a pivotal task in natural language processing, bridging the gap between unstructured user queries and structured tabular data. Recent advancements in Large Language Models (LLMs) have significantly enhanced TableQA performance, particularly through the development of agent-based paradigms. However, existing agent-based frameworks often suffer from a critical “self-verification bias,” as they tend to uphold their own erroneous reasoning paths rather than objectively identifying logical pitfalls, leading to error propagation. To address this issue, we propose a novel iterative multi-agent framework that breaks this bias by substituting unreliable self-correction with an independent, objective auditing mechanism. Specifically, our framework consists of a Reasoner Agent, an Analyst Agent, and an Arbiter Agent. To explicitly overcome the verification bottleneck, the Analyst Agent is fine-tuned via adversarial critique distillation, enabling it to generate precise critiques and error localizations. During inference, instead of relying on intrinsic self-reflection, the framework utilizes the explicit critiques from the Analyst Agent as external feedback to iteratively refine the parallel generation steps of the Reasoner Agent, effectively halting error propagation. Finally, the Arbiter Agent employs a hierarchical arbitration to derive the optimal consensus. Experiments on multiple benchmarks demonstrate that our framework achieves significant performance improvements over existing baselines, validating its effectiveness in overcoming intrinsic self-correction limitations.

Keywords: Table Question Answering · Large Language Models · Multi-Agent Framework · Adversarial Critique Distillation.

1 Introduction

Table Question Answering (TableQA) stands as a pivotal task in natural language processing, bridging the gap between unstructured natural language queries

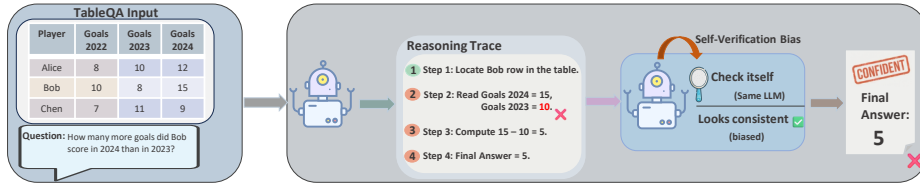


Fig. 1. The LLM checks its own flawed reasoning and still outputs a confident wrong answer.

and structured tabular data [31, 6, 23]. As tables are a ubiquitous format for storing relational information in scientific reports, financial statements, and web documents, the ability to perform accurate information retrieval and multi-step reasoning over these structures possesses significant academic and industrial value, and has thus been widely studied [33]. Recent advancements in Large Language Models (LLMs) have significantly enhanced TableQA performance, particularly through the development of agent-based paradigms [38, 10, 2]. These methods typically decompose complex questions into executable sub-steps or iterative reasoning chains, performing step-by-step inference followed by autonomous self-verification or reflection to ensure the logical consistency of the derived answers.

However, a challenge in current agent-based TableQA methods: the self-verification bias. While the self-verification phase aims to identify errors to ensure logical consistency, it often fails in practice because LLMs tend to generate highly confident but erroneous refinements during self-correction that merely reiterate their initial misconceptions [36]. This phenomenon leads to a self-reinforcing bias, where the model justifies its erroneous reasoning steps rather than objectively identifying logical pitfalls, thereby causing error propagation. As illustrated in Fig. 1, the self-verification stage can reinforce an initially flawed reasoning trace: the model checks its own steps with the same underlying bias and still outputs a highly confident but incorrect answer, causing error propagation in TableQA.

To address this issue, we train a specialized Analyst Agent using contrastive reasoning supervision derived from existing TableQA datasets. Specifically, we construct the training data from the training splits of WikiTableQuestions[19] and TabFact [1]. For each instance, we generate reasoning trajectories using two different large language models, namely DeepSeek-V3 [15] and Qwen3-8B[24]. We then identify cases where the two models produce different outcomes, particularly when one model arrives at the correct answer while the other produces a plausible but logically flawed reasoning path. These cases naturally form contrastive positive-negative reasoning pairs consisting of valid and erroneous reasoning trajectories. The Analyst Agent is trained on these pairs to learn structural patterns that distinguish correct reasoning from flawed reasoning and to localize errors within the reasoning process.

Based on this Analyst Agent, we further develop an iterative multi-agent framework that integrates a Reasoner Agent, the Analyst Agent, and an Arbitrer

Agent (**RNA**). Instead of relying on the model’s own self-correction, the Analyst functions as an independent verifier that critiques the reasoning trajectory and identifies potential errors, while the Arbiter selects the most reliable answer among candidate solutions. This collaborative process mitigates self-verification bias by replacing unreliable self-validation with an external auditing mechanism. Specifically, during inference, the framework follows an iterative refinement process with parallel reasoning paths. For each query, the Reasoner Agent first generates multiple candidate reasoning trajectories based on the input table and question. The Analyst Agent then independently evaluates these trajectories and provides explicit critiques that highlight potential logical inconsistencies. Guided by this feedback, the Reasoner Agent revises its reasoning in subsequent iterations. Empirically, we set the maximum number of iterations to $K=3$, which provides a favorable trade-off between computational cost and performance improvement. Finally, the Arbiter Agent performs hierarchical arbitration over the candidate reasoning paths to select the most reliable answer, ensuring the robustness of the final prediction.

We evaluate our framework on common datasets. Experimental results demonstrate that our multi-agent collaboration mechanism achieves significant improvements compared to the base model. Furthermore, our ablation studies confirm that the Analyst Agent, empowered by adversarial distillation, provides a substantially stronger verification signal than the model’s intrinsic self-correction. These findings confirm that, in complex tabular reasoning tasks, independent and specialized auditors are essential for overcoming the limitations of intrinsic self-correction. Our contributions can be summarized as follows:

- We identify and formally characterize the phenomenon of *self-verification bias* in agent-based TableQA systems, where models tend to validate their own reasoning trajectories even when logical errors occur.
- We introduce an adversarial contrastive training strategy for the Analyst Agent, constructed from cross-model reasoning disagreements. By leveraging paired correct and erroneous reasoning trajectories, the Analyst learns to detect structural inconsistencies in reasoning and provide targeted critiques.
- We incorporate the trained Analyst into an iterative reasoning framework with a Reasoner Agent and an Arbiter Agent, forming an independent auditing mechanism that mitigates error propagation. Experiments on WTQ and TABFACT demonstrate consistent improvements in reasoning reliability and answer accuracy.

2 Related Work

2.1 Table Question Answering

Table Question Answering (TableQA) has evolved from semantic parsing systems to neural end-to-end reasoning models. Early approaches primarily relied on weak supervision and symbolic semantic parsing, translating natural language

questions into executable programs such as SQL to enable structured reasoning over tables [17, 20].

With the emergence of large language models (LLMs), recent work has shifted toward leveraging pre-trained models for tabular reasoning. These approaches improve schema linking, evidence retrieval, and representation of semi-structured data, enabling more flexible reasoning across heterogeneous tables [18, 12, 5]. Despite these advances, complex TableQA tasks often require multi-step reasoning, making reliable reasoning verification an increasingly important challenge.

2.2 Agent-based and Multi-Agent TableQA

Recent studies increasingly model LLMs as reasoning agents capable of decomposing complex questions into intermediate steps. These agent-based approaches typically adopt plan-then-execute paradigms to perform structured reasoning and tool usage [9, 7, 13].

Building on this idea, multi-agent frameworks introduce role specialization among multiple LLM instances, allowing different agents to collaboratively generate, refine, or validate candidate solutions [25, 14]. Some approaches also integrate external tools such as SQL executors or retrieval systems to further improve reasoning accuracy [21, 30].

While these frameworks improve task decomposition and reasoning diversity, their verification mechanisms are often limited to answer-level checks or consensus-based filtering. When multiple agents share similar model architectures or prompting strategies, they may reinforce one another’s errors rather than provide effective critique. This limitation highlights the need for independent auditing mechanisms that can evaluate reasoning paths more objectively.

2.3 Reasoning Verification and Self-Reflection

Improving the reliability of LLM reasoning has become an active research direction. Existing approaches attempt to enhance reasoning quality through process-based preference learning [37] or by improving the coherence and stability of intermediate reasoning steps [34]. Other studies explore self-reflection mechanisms in which models iteratively evaluate and revise their own reasoning outputs [11].

However, these approaches typically rely on the same model to both generate and evaluate reasoning trajectories. As a result, the verification process remains tightly coupled with the generator’s internal biases. In practice, models often tend to defend their initial reasoning assumptions during self-reflection, even when those assumptions are incorrect. This phenomenon leads to what we refer to as *self-verification bias*, where a model preferentially validates its own reasoning trajectory despite the presence of logical errors.

To address this limitation, our work introduces an independent auditing mechanism that separates reasoning generation from reasoning verification. By training a dedicated Analyst Agent on contrastive reasoning trajectories, the framework enables more reliable identification of flawed reasoning paths and supports iterative correction during inference.

3 Problem Definition and Preliminaries

In this section, we introduce the formulation of agent-based Table Question Answering (TableQA) and formally define the notion of *self-verification bias*.

3.1 Agent-based TableQA

Table Question Answering (TableQA) aims to derive a concise answer from a semi-structured table and a natural-language question. Let T denote a table represented in serialized textual form and q denote the input question. A TableQA system produces an answer a , which can be expressed as the mapping

$$f : (T, q) \mapsto a. \quad (1)$$

Recent TableQA approaches increasingly adopt agent-based reasoning frameworks. Instead of directly predicting an answer, the system generates intermediate reasoning steps that describe how the answer is derived from the table. We denote such a reasoning trajectory as a *reasoning path*

$$r = (s_1, s_2, \dots, s_m), \quad (2)$$

where each s_j represents an intermediate reasoning step.

Given a reasoning path r , the final answer is extracted through a deterministic function

$$a = \text{Extract}(r). \quad (3)$$

3.2 Verification Process

To ensure correctness, many agent-based frameworks introduce a verification stage that evaluates whether a reasoning path is valid with respect to the table and question. Formally, a verifier can be represented as a function

$$V(T, q, r) \in \{0, 1\}, \quad (4)$$

where $V(T, q, r) = 1$ indicates that the reasoning path r is considered valid and $V(T, q, r) = 0$ indicates that the path is rejected.

In many existing systems, both reasoning and verification are performed by the same language model or by models with highly similar training distributions.

3.3 Self-Verification Bias

When the reasoning generator and the verifier share similar generation patterns, the verification outcome may become correlated with the generation process itself rather than with the actual correctness of the reasoning.

Definition 1 (Self-Verification Bias). Let M_g denote a reasoning generator that produces a reasoning path r , and let M_v denote a verifier that evaluates the

path using the function $V(T, q, r)$. Self-verification bias occurs when M_g and M_v share similar model priors such that

$$P(V(T, q, r) = 1 \mid r) \quad (5)$$

is influenced more by the generative patterns of M_g than by the factual correctness of r with respect to (T, q) .

In iterative reasoning settings, where verification outcomes are used to guide subsequent reasoning steps, such bias may repeatedly reinforce incorrect reasoning paths, resulting in cascading error propagation during inference.

4 Method

To mitigate the self-verification bias described in Section 3, we propose RNA, a multi-agent reasoning framework for TableQA. RNA explicitly separates candidate reasoning generation, verification, and final answer selection through three specialized agents: the Reasoner Agent ($\mathcal{R}$), the Analyst Agent ($\mathcal{N}$), and the Arbiter Agent ($\mathcal{A}$). Figure 2 provides an overview of the framework.

4.1 Adversarial Critique Training of the Analyst Agent

The Analyst Agent $\mathcal{N}$ is responsible for evaluating reasoning paths produced by the Reasoner Agent. To enable this, we construct an *adversarial critique distillation* dataset that provides both positive and negative reasoning examples.

Formally, the dataset is

$$\mathcal{D}_{\text{acd}} = \{(T, q, r, y^*)\}, \quad (6)$$

where T is the serialized table, q is the question, r is a reasoning path, and y^* is the teacher-generated critique target.

We independently generate reasoning trajectories for each training instance using two distinct large language models, DeepSeek-V3 [15] and Qwen3-8B [24]. We identify cases where the two models produce different outcomes, specifically when one model derives the correct answer while the other generates a plausible but logically flawed reasoning path. These natural disagreements form high-quality contrastive pairs, where the correct trajectory serves as the positive example and the flawed one as the negative example. This approach naturally captures a wide range of typical reasoning errors, including missing steps, wrong arithmetic operations, and misaligned table cells, allowing the Analyst Agent to learn fine-grained error detection capabilities.

The critique target for each reasoning path is

$$y^* = \langle v^*, e^* \rangle, \quad (7)$$

where $v^* \in \{0, 1\}$ indicates validity, and e^* provides a structured description of the error if invalid. By including localized error information, the Analyst Agent

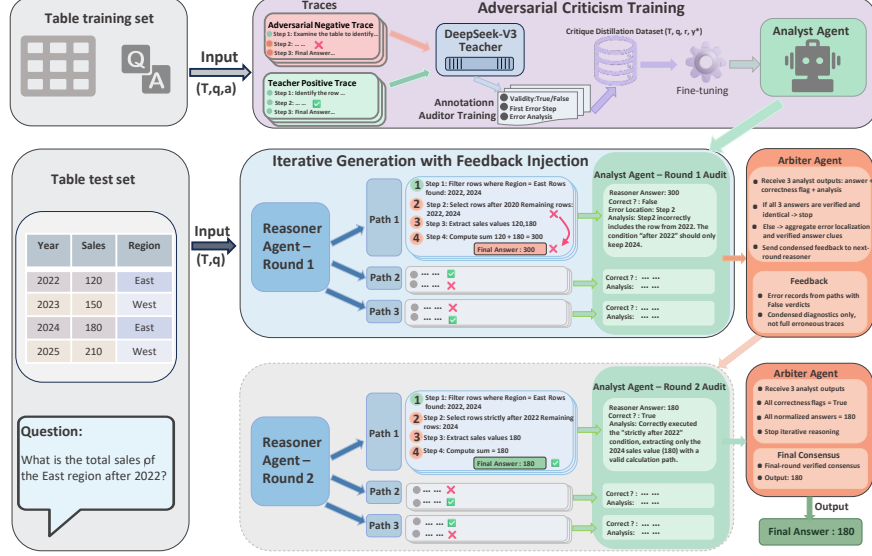


Fig. 2. Overview of RNA multi-agent framework for TableQA. The Analyst Agent is first trained via adversarial critique distillation. During inference, the Reasoner Agent iteratively generates candidate reasoning paths, audited by the Analyst Agent. Feedback signals guide subsequent reasoning rounds. Finally, the Arbiter Agent performs hierarchical arbitration to select the final answer.

can not only verify correctness but also generate actionable feedback for iterative reasoning.

The Analyst Agent is trained via supervised fine-tuning:

$$\mathcal{L}(\theta_{\mathcal{N}}) = -\mathbb{E}_{(T,q,r,y^*) \sim \mathcal{D}_{\text{acd}}} [\log P_{\theta_{\mathcal{N}}}(y^* | T, q, r)]. \quad (8)$$

After training, given a reasoning path r , the Analyst Agent outputs

$$c = \langle v, e \rangle, \quad (9)$$

where v is the verification result and e contains the localized error description.

4.2 Agent Definitions and Roles

Reasoner Agent $\mathcal{R}$ generates candidate reasoning paths conditioned on the input table-question pair (T, q) and optionally on feedback from previous reasoning rounds. Its role is to explore diverse solution strategies, including multi-hop reasoning and aggregation across table cells.

Analyst Agent $\mathcal{N}$ serves as an independent verifier and error localizer. It evaluates reasoning paths, identifies specific mistakes, and produces structured feedback signals. This feedback is crucial for iterative improvement, enabling the Reasoner Agent to avoid repeating known errors.

Arbiter Agent $\mathcal{A}$ consolidates verified candidate answers and determines the final prediction. It employs a hierarchical arbitration strategy that balances reliance on the most recent reasoning outputs with robustness across the entire reasoning history.

4.3 Iterative Reasoning with Feedback Injection

Inference proceeds iteratively for up to K rounds. At round t , the Reasoner Agent generates N candidate reasoning paths:

$$\mathcal{P}_t = \{r_t^{(1)}, r_t^{(2)}, \dots, r_t^{(N)}\}. \quad (10)$$

Each path yields a candidate answer:

$$a_t^{(i)} = \text{Extract}(r_t^{(i)}), \quad (11)$$

forming the answer set

$$\mathcal{A}_t = \{a_t^{(1)}, a_t^{(2)}, \dots, a_t^{(N)}\}. \quad (12)$$

The Analyst Agent evaluates each path:

$$c_t^{(i)} = \langle v_t^{(i)}, e_t^{(i)} \rangle, \quad (13)$$

producing the critique set

$$\mathcal{C}_t = \{c_t^{(1)}, c_t^{(2)}, \dots, c_t^{(N)}\}. \quad (14)$$

If all paths are verified and candidate answers are consistent, the process terminates early. Otherwise, a feedback signal aggregates error diagnostics and reference answers:

$$F_t = \text{Agg} \left(\{e_t^{(i)} \mid v_t^{(i)} = 0\} \cup \{a_t^{(i)} \mid v_t^{(i)} = 1\} \right), \quad (15)$$

where $\text{Agg}(\cdot)$ condenses multiple errors and verified candidates into actionable guidance for the next round.

The Reasoner Agent then generates the next candidate set conditioned on the feedback:

$$r_{t+1}^{(i)} \sim P_{\mathcal{R}}(\cdot \mid T, q, F_t). \quad (16)$$

This iterative refinement allows the system to recover from initial mistakes, progressively improving reasoning accuracy.

4.4 Hierarchical Arbitration by the Arbiter Agent

After termination at round $K' \leq K$, the Arbiter Agent aggregates all candidate answers:

$$\mathcal{A}_{\leq K'} = \bigcup_{t=1}^{K'} \mathcal{A}_t. \quad (17)$$

Answers are normalized:

$$\tilde{a} = \text{Norm}(a), \quad (18)$$

to reduce sensitivity to surface-form variations.

Hierarchical arbitration proceeds as follows:

1. ****Final-round verified candidates****: if any exist, majority voting determines the final answer.
2. ****Backtracking****: if no verified paths exist in the last round, aggregate all verified answers across previous rounds.
3. ****Global fallback****: if no verified paths exist at all, perform majority vote over all generated candidates.

The final prediction is

$$\hat{a} = \text{Vote}(\tilde{\mathcal{A}}), \quad (19)$$

where $\tilde{\mathcal{A}}$ is the normalized candidate set selected by the arbitration procedure.

5 Experiments

5.1 Experimental Setup

Datasets and Evaluation. We evaluate our framework on two established benchmarks for tabular question answering and fact verification: WikiTableQuestions (WTQ) [19] and TabFact [1]. WTQ involves complex compositional reasoning over semi-structured tables, requiring operations like aggregation, comparison, and arithmetic calculations to derive concise answers. We use the official test split of 4,344 examples, with performance measured by the Exact Match (EM) metric [16]. TabFact is a fact verification dataset comprising 2,024 test instances, where the system must determine if a given statement is entailed or refuted by the provided table. For this task, we report classification accuracy.

Base Models and Thinking Mode. We use Llama3.1-8B[4] and Qwen3-8B[24] as the base reasoners in our experiments. For Qwen3-8B, we further consider two settings of its built-in *Thinking Mode*, with Thinking Mode disabled / enabled. For Llama3.1-8B, this setting is not applicable and is denoted as N/A. Our experiments follow the framework described in Section 4, where the Analyst Agent is trained separately via adversarial critique distillation and the Arbiter Agent performs the hierarchical arbitration procedure.

Baselines. To contextualize the empirical performance of our framework, we compare against three groups of methods. The first group consists of prior specialized, pipeline-based methods, including Table-Critic [29], CIT [27], BINDER [3], StructGPT [8], DATER [28], CHAIN-OF-TABLE [22], ReAcTable [35], SYN-TQA [32], and TIDE [26]. The second group consists of general-purpose LLMs that we evaluate under a unified single-pass Chain-of-Thought prompting setup. The third group consists of the same base models integrated with our framework, each compared against their corresponding single-pass CoT baselines ([†]).

Implementation Details. For framework integration, we set the number of parallel candidate reasoning paths to $N = 3$ and the maximum number of refinement rounds to $K = 3$. Candidate paths are generated via stochastic

Table 1. Main results on the WikiTable-Questions (WTQ) test set.

Method / Model	EM. (%)
<i>Specialized / Pipeline Methods</i>	
BINDER (ICLR'23)	64.60
DATER (SIGIR'23)	65.90
StructGPT (EMNLP'23)	57.00
CHAIN-OF-TABLE (ICLR'24)	59.94
ReAcTable (VLDB'24)	68.00
SYNTQA (EMNLP'24)	70.40
Table-Critic (ACL'25)	73.90
TIDE (ICLR'25)	75.00
CIT (NeurIPS'25)	76.38
<i>General LLMs (Single-pass CoT)</i>	
GPT-4o	64.39
GPT-4.1	68.00
DeepSeek-V3	69.90
Qwen2.5-32B	<u>77.30</u>
<i>Ours</i>	
Llama3.1-8B	55.23
Llama3.1-8B (+RNA)	62.10 (+6.87)
Qwen3-8B	57.54
Qwen3-8B (+RNA)	74.18 (+16.64)
Qwen3-8B (Thinking Mode)	73.02
Qwen3-8B (Thinking Mode) (+RNA)	80.04 (+7.02)

Table 2. Main results on the TabFact test set.

Method / Model	EM. (%)
<i>Specialized / Pipeline Methods</i>	
BINDER (ICLR'23)	85.10
DATER (SIGIR'23)	85.60
StructGPT (EMNLP'23)	87.30
CHAIN-OF-TABLE (ICLR'24)	80.20
ReAcTable (VLDB'24)	86.10
TIDE (ICLR'25)	89.82
Table-Critic (ACL'25)	91.10
Weaver (EMNLP'25)	91.20
CIT (NeurIPS'25)	91.30
<i>General LLMs (Single-pass CoT)</i>	
GPT-4o	73.33
GPT-4.1	86.50
DeepSeek-V3	<u>92.40</u>
Qwen2.5-32B	90.30
<i>Ours</i>	
Llama3.1-8B	82.16
Llama3.1-8B (+RNA)	83.70 (+1.54)
Qwen3-8B	85.72
Qwen3-8B (+RNA)	92.14 (+6.42)
Qwen3-8B (Thinking Mode)	90.03
Qwen3-8B (Thinking Mode) (+RNA)	93.03 (+3.00)

decoding with temperature set to 0.3. Early stopping is triggered when all candidate paths in the current round are verified as valid and yield an identical answer.

5.2 Main Results

Table 1 and Table 2 summarize the main results on WTQ and TabFact, respectively. Overall, integrating the proposed RNA framework consistently enhances performance across both datasets, demonstrating its effectiveness in improving reasoning quality and answer verification.

Performance on WTQ. On WTQ, the base models exhibit moderate performance under single-pass CoT prompting (55.23% for Llama3.1-8B and 57.54% for Qwen3-8B). Incorporating the framework yields substantial gains, particularly for Qwen3-8B (+16.64), indicating that the iterative reasoning with Analyst feedback and hierarchical arbitration effectively corrects errors that single-pass generation misses. Even when Qwen3-8B operates in Thinking Mode, which already boosts its baseline to 73.02%, the framework further increases accuracy to 80.04% (+7.02), surpassing specialized pipelines and approaching or exceeding the performance of strong general-purpose LLMs. These observations suggest that the framework’s verification and feedback mechanism systematically improves reasoning robustness rather than simply amplifying the model’s existing strengths.

Performance on TabFact. A consistent improvement pattern is observed on TabFact. Llama3.1-8B gains 1.54 points and Qwen3-8B improves by 6.42 points, while enabling Thinking Mode further elevates Qwen3-8B by an additional 3 points. The relatively larger improvement for Qwen3-8B highlights that models with stronger base reasoning benefit more from the multi-agent framework, likely because the Analyst Agent can more effectively audit and refine high-quality candidate paths. Overall, the results indicate that the RNA framework provides meaningful enhancements over both weaker and stronger base models, yielding final predictions that are more reliable and robust than single-pass or pipeline baselines.

5.3 Ablation Study

To evaluate the contribution of the independently trained Analyst Agent and the multi-agent design, we conduct ablation experiments with two variants: *w/o Adversarial Critique Training*, where the Analyst Agent is replaced by self-auditing of the Reasoner Agent, and *w/o Multi-agent Framework*, where both the Analyst and Arbiter agents are removed and only single-pass reasoning is used. All other components, such as feedback injection (where applicable), are retained for fair comparison.

Table 3. Ablation study results. Values in parentheses indicate absolute performance drop (↓) relative to the full method.

Method	WTQ	TabFact
Llama3.1-8B	62.10	83.70
w/o Adversarial Critique Training	56.62 (↓ 5.48)	80.39 (↓ 3.31)
w/o Multi-agent Framework	55.23 (↓ 6.87)	82.16 (↓ 1.54)
Qwen3-8B	74.18	92.14
w/o Adversarial Critique Training	67.88 (↓ 6.30)	87.26 (↓ 4.88)
w/o Multi-agent Framework	57.54 (↓ 16.64)	85.72 (↓ 6.42)
Qwen3-8B (Thinking Mode)	80.04	93.03
w/o Adversarial Critique Training	76.43 (↓ 3.61)	90.36 (↓ 2.67)
w/o Multi-agent Framework	73.02 (↓ 7.02)	90.03 (↓ 3.00)

Table 3 highlights the importance of both adversarial critique training and the multi-agent framework. Replacing the Analyst Agent with self-auditing (*w/o Adversarial Critique Training*) leads to noticeable performance drops across both datasets, ranging from 3.61 to 6.30 points, indicating that independent critique is crucial for correcting errors that the Reasoner Agent alone cannot detect. Removing the multi-agent framework entirely (*w/o Multi-agent Framework*) results in even larger degradations, particularly for Qwen3-8B without Thinking Mode, where the drop reaches 16.64 points on WTQ.

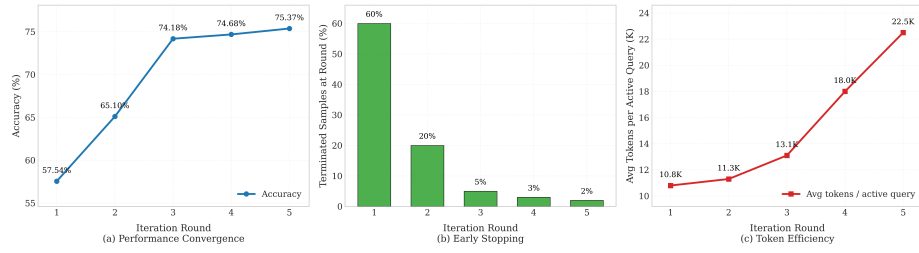


Fig. 3. In-depth analysis of the iterative refinement process on the WTQ dataset: (a) Accuracy growth across rounds showing diminishing returns after $K = 3$; (b) Discrete distribution of samples triggering early stopping per round; (c) Average token consumption per active query, illustrating the disproportionate cost in later stages.

These results provide two key insights. First, self-verification is insufficient: even strong base models benefit from an external Analyst Agent to mitigate self-verification bias. Second, the multi-agent architecture, combining iterative reasoning, feedback injection, and hierarchical arbitration, significantly enhances model robustness and answer accuracy, particularly on complex table reasoning tasks. Overall, the ablation study confirms that both adversarial critique training and the multi-agent design are essential components for achieving the improvements reported in the main results.

5.4 Experiments Analysis

In this section, we analyze the iterative behavior of the proposed framework in order to understand the effectiveness of the refinement process and to justify the choice of the maximum iteration number $K = 3$. All analysis experiments are conducted on the WTQ test set using Qwen3-8B (Thinking Mode disabled), which corresponds to the primary experimental setting in our main results. We specifically select Qwen3-8B for this ablation as it represents a highly capable base model; analyzing its iterative dynamics provides profound insights into how our framework benefits models that already possess strong foundational reasoning capabilities, thereby isolating the performance gains directly attributable to our multi-agent critique mechanism.

Performance Convergence and Iterative Gain. We first evaluate how the model performance evolves as the number of iterations increases. As shown in Fig. 3(a), the initial accuracy (Round 1) starts at 57.54%. With the feedback provided by the Analyst Agent, we observe a significant performance jump in the subsequent two rounds, reaching 65.1% and 74.18%, respectively. However, the improvement becomes much smaller afterward, increasing only to 74.68% at Round 4 and 75.37% at Round 5. These results suggest that three rounds are sufficient to capture the major benefits of iterative refinement, while additional rounds contribute only marginal improvements. We hypothesize that the

performance plateau observed after the third iteration is largely due to inherent dataset constraints, such as unresolvable ambiguities in the table schema, missing information, or questions that require domain knowledge strictly outside the provided context. For these intrinsically unanswerable or highly ambiguous queries, no amount of iterative refinement can logically derive the correct answer, thus making further iterations futile.

Early Stopping Behavior. The efficiency of our framework is largely driven by the early stopping mechanism, which terminates the process once the Analyst Agent verifies all parallel paths as consistent and correct. Fig. 3(b) illustrates the discrete distribution of samples satisfying the termination criteria at each specific round. We observe that 60% of the queries are resolved and terminate in the very first round. The volume of samples necessitating further iterative refinement tapers off rapidly in subsequent stages, with only 20% and 5% of the dataset terminating in Round 2 and Round 3, respectively. The termination rates for the final rounds remain marginal, accounting for a mere 3% in Round 4 and 2% in Round 5. This distribution confirms that $K = 3$ covers the vast majority of cases, as increasing the iteration limit beyond this point would only impact a negligible fraction of the dataset while unnecessarily raising the overall computational complexity.

Computational Cost and Token Efficiency. To further validate the economic rationale of $K = 3$, we analyze the cumulative token consumption across iterations. As depicted in Fig. 3(c), the token usage grows linearly during the first three rounds, corresponding to the steady refinement of reasoning paths. Notably, for the few samples that proceed to Round 4 and Round 5, we observe a disproportionate increase in token consumption (a steeper slope), likely because the Reasoner generates more verbose trajectories when struggling with persistent errors. Specifically, our empirical analysis reveals that the average token consumption per active query increases from 10.8K in Round 1, to 11.3K in Round 2, and 13.1K in Round 3. For the final 5% of samples that persist into Round 4 and Round 5, this cost escalates sharply to 18.0K and 22.5K per query, respectively. This confirms that struggling models tend to generate more verbose and redundant reasoning paths when unable to resolve a query, making further iterations after $K = 3$ economically inefficient.

Overall, the three analyses consistently support the design choice of setting the maximum refinement depth to $K = 3$. At this point, the framework captures the majority of accuracy gains, most samples have already terminated through early stopping, and the computational cost remains manageable. Increasing the iteration budget beyond this point leads to significantly higher inference cost while yielding only marginal improvements in answer accuracy.

6 Limitations

While our framework is designed to mitigate self-verification bias, it still has several limitations. First, our evaluation is restricted to structured tabular reasoning

on WTQ and TabFact with 8B-parameter open-weight models. Although this setup is cost-effective, broader experiments on additional TableQA benchmarks and larger models (e.g., 70B+) are needed to determine how well the framework generalizes and whether self-verification bias diminishes at scale. Second, the Analyst Agent is a learned critic rather than an oracle: its critiques may still be imperfect and, in difficult cases, may mislead subsequent refinement rounds. Third, the current Analyst relies purely on learned internal patterns through adversarial critique distillation, without incorporating external symbolic executors (e.g., Python or SQL) or retrieval systems. Extending the Analyst with tool-assisted verification is an important direction for improving robustness, especially beyond structured tabular reasoning.

7 Conclusion

We presented an iterative multi-agent framework for TableQA that addresses the self-verification bias of existing agent-based reasoning systems by replacing unreliable self-correction with independent external auditing. Our framework decouples candidate reasoning generation, fine-grained critique, and final consensus selection through a Reasoner Agent, an Analyst Agent trained with adversarial critique distillation, and an Arbiter Agent with hierarchical arbitration. Experiments on WTQ and TabFact show that this design consistently improves the underlying base models, while ablations confirm that independent critique is substantially more effective than self-auditing. Overall, our results suggest that, for complex tabular reasoning, reliable performance depends not only on stronger generation but also on explicitly separating reasoning from verification.

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